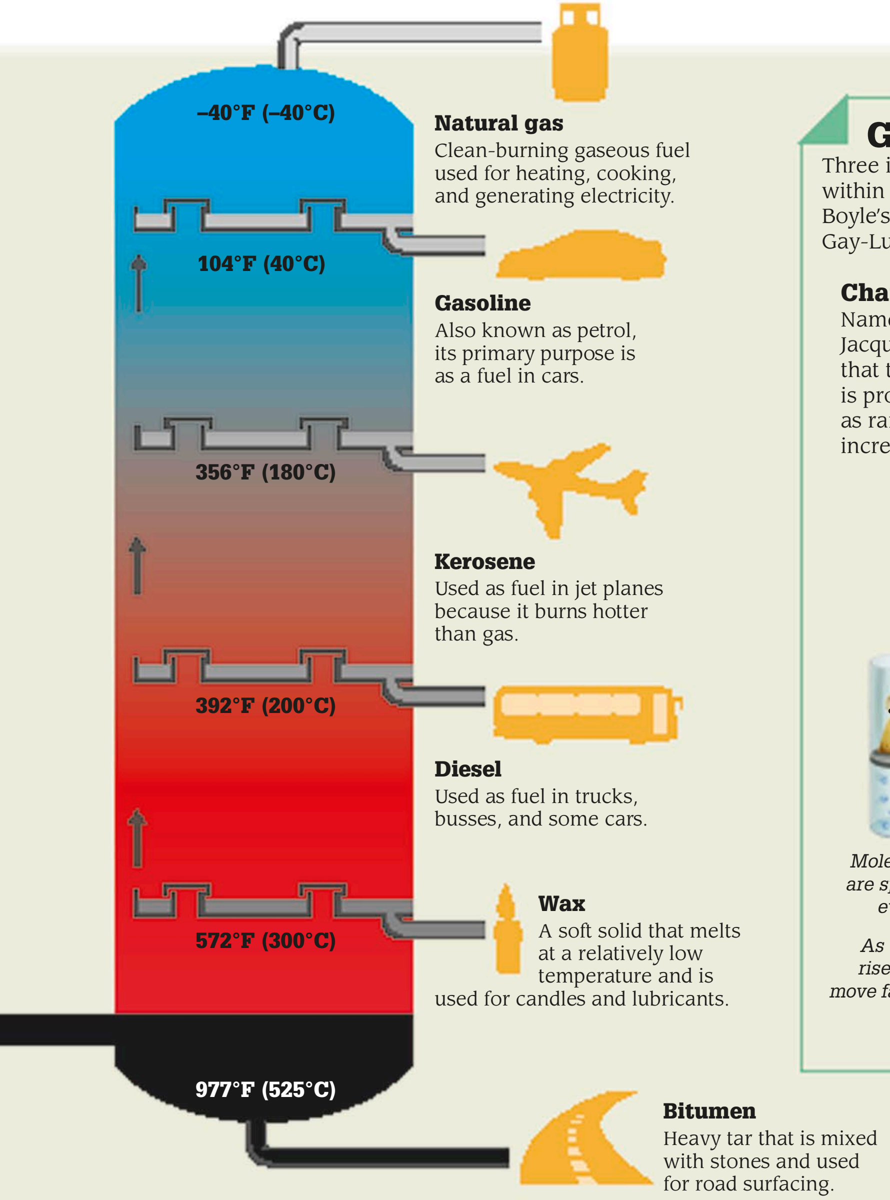


Chemistry in industry

As research into chemical processes progressed, industry began to use the findings to develop the processes we see today. One such process is the distillation of crude oil—a complex mixture of hydrocarbons (compounds of hydrogen and carbon). These are separated inside a fractionating column (above). As the oil boils, the hydrocarbons turn to gas and rise up the tower. They rise, then cool and condense (turn back to liquid) at different temperatures, and are collected and refined into useful products, such as gas.

Gas laws

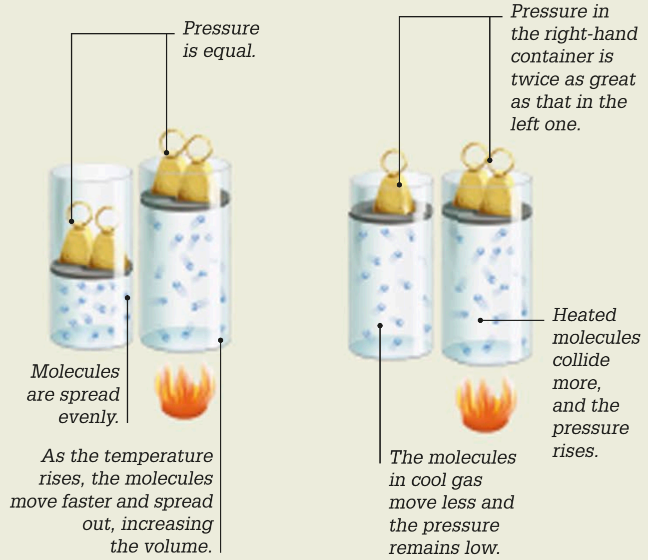
Three important laws describing the movements of molecules within gases are named after their discoverers. The first is Boyle's law (see p.83). The other two are Charles' law and Gay-Lussac's law.

Charles' law

Named after French scientist Jacques Charles, this law states that the temperature of a gas is proportional to its volume, as raising the temperature increases the volume.

Gay-Lussac's law

First described by French chemist Joseph Louis Gay-Lussac, this law states that for a fixed volume of gas, the pressure is proportional to its temperature.



Synthetic polymers

Plastics were first made in the late 1800s. Produced from organic substances such as wood, coal, or gas, they are made of polymers—large molecules that are molded into shape when heated.

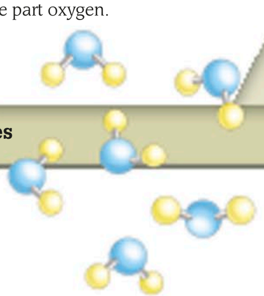
Dice made of celluloid, one of the first plastics

1803

English chemist John Dalton presented his atomic theory—the first attempt to describe all matter in terms of atoms and their properties.

1805

Joseph Louis Gay-Lussac discovered that water is made up of two parts hydrogen and one part oxygen.



Water molecules

1811

Italian physicist Amedeo Avogadro realized that simple gases such as hydrogen are made up of molecules of two or more atoms joined together.

1869

Dmitri Mendeleev published the periodic table in which he organized the 66 elements that were known at the time by their atomic weights and properties.